

Deepak Mallampati

(330) 554-0770 | deepakmallampati00@gmail.com | linkedin.com/in/deepak-mallampati | github.com/deepak-glitch |
US-based, F-1 OPT, open to sponsorship | Open to NYC relocation

PROFESSIONAL SUMMARY

Applied AI engineer (2.5+ years) shipping production RAG, agentic LLM workflows, and evaluation pipelines in HIPAA-conscious healthcare. Comfortable bridging technical depth with stakeholder narratives - ~35% retrieval precision gain, >30% hallucination reduction, >90% grounded response alignment, and ~30% fewer post-deploy defects. Hands-on with LangChain, LlamaIndex, FastAPI, Docker, and multi-agent orchestration. Looking to apply this build-and-evaluate discipline to AI Solutions Engineering, where the customer is the next teammate on the system.

CORE COMPETENCIES

RAG Engineering • Agentic LLM Workflows • LLM Evaluation & Observability • Grounding & Guardrails • LangChain • LlamaIndex • LangGraph (adjacent) • Python • FastAPI • Flask • Docker • REST APIs • scikit-learn • XGBoost • PyTorch • Hugging Face • SQL (T-SQL, PostgreSQL) • TypeScript / React • MLOps / LLMOps • Customer-facing Stakeholder Docs • HIPAA-conscious Data Governance

PROFESSIONAL EXPERIENCE

Jr. AI/ML Engineer Trainee — Progress Solutions Inc. *Jul 2025 – Present* | USA | Healthcare AI / SaaS

- Built **Retrieval-Augmented Generation (RAG) systems** for clinical knowledge retrieval; recursive semantic chunking + transformer embeddings improved contextual retrieval precision ~**35%** and reduced irrelevant retrieval >**30%**; grounded response alignment >**90%** in evaluation.
- Maintained **evaluation dashboards and stakeholder-facing system-limitation reports** that surfaced retrieval drift and hallucination patterns for downstream care teams — the customer-facing artifact behind every production deploy.
- Developed **agentic LLM workflows** for multi-step healthcare queries with structured reasoning, tool discipline, and grounding rules; improved agent response stability ~**25%**.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient no-show, care engagement, and support prioritization; recall +**15–20%** on high-risk categories at precision >**90%**.
- Packaged ML/LLM inference as **FastAPI/Flask** RESTful services, containerized with **Docker**; reduced post-deployment defects ~**30%**.
- Built preprocessing pipelines (Pandas, NumPy) for EHR extracts, appointment histories, support ticket logs; raised dataset reliability >**98%** and cut downstream model instability ~**40%**.

Database & DevOps Performance Engineer (Intern) — Energy Solutions International (Emerson) *Jun 2022 – Apr 2023* | Hyderabad, India | Enterprise ERP

- Optimized **T-SQL stored procedures** powering the Synthesis Order-to-Cash platform; cut query execution ~**20%** and data retrieval latency ~**25%**.
- Built **CI/CD pipelines with Jenkins** for schema updates and stored procedure deployments; reduced deployment errors >**30%** and improved release cycle time ~**35–40%**.
- Designed performance dashboards (SQL Server DMVs, Grafana) for long-running queries, deadlocks, CPU/I/O contention; cut incident recurrence ~**25%**.

Machine Learning Engineer — Suvidha Foundation *Jan 2022 – Mar 2022* | Hyderabad, India

- Built a **transformer-based video summarization** system across 5,000+ recorded sessions; reduced manual review time **60–70%**.
- Implemented **RAG-based document Q&A**; reduced hallucinated outputs ~**30%**; deployed as **Flask API**.

SELECTED PROJECTS

Manga Lens — Multi-Provider AI Vision Chrome Extension (*shipped, Chrome Web Store*)

Solo-built Chrome extension integrating **four AI vision providers** (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama). Multi-section capture pipeline, per-domain selector configs for 29 sites, multi-provider payload routing (WebP for cloud, JPEG for Ollama). Direct customer feedback loop via Chrome Web Store reviews shaped multi-provider fallback — same instinct an SE applies to enterprise customer signals.

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

Multi-agent pipeline (Intake → Validation → Consistency → Duplicate Detection → Fraud Risk) with **schema-validated JSON contracts between agents** to prevent cascading hallucinations — same problem class Arize observability customers face. RAG-grounded CPT/ICD validation; explainable risk scoring with audit-ready reasoning traces.

Dream Decoder — Multimodal Creative Intelligence Platform

FastAPI + React/TypeScript app with intermediate **structured prompt transformation layers**; improved contextual alignment ~**30%** and first-pass image success ~**25–30%**.

Driver Drowsiness Detection — Real-Time YOLOv8

Unified YOLOv8 detection + classification, sliding-window confidence aggregation; ~**30%** latency reduction, ~**25%** fewer false positives.

EDUCATION

Master's Degree — Kent State University (*Aug 2023 – May 2025*) | *Ohio, USA*

CERTIFICATIONS

Available upon request

TECHNICAL SKILLS

Languages & Frameworks: Python, FastAPI, Flask, SQL (T-SQL, PostgreSQL, SQLite), TypeScript, React, C++ (Arduino), PHP

LLM & GenAI: RAG, agentic workflows, structured outputs, prompt engineering, evaluation pipelines (precision/recall, grounded-answer ratio), guardrails, grounding, embeddings, vector search, LLM observability

ML Libraries: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, Diffusers, LangChain, LlamaIndex

Computer Vision: YOLOv8, OpenCV, ControlNet, OpenPose/MediaPipe, Stable Diffusion

Data: Pandas, NumPy, feature engineering, time-series forecasting, walk-forward validation

Infra & DevOps: Docker, Jenkins, Grafana, CI/CD, MLOps/LLMOps, RESTful APIs, observability, structured logging

Domains: Healthcare AI (HIPAA-conscious), LLM observability, Applied AI, Multimodal Systems